Instructions:
You are a SOC Analyst at Meoware Labs. Over the past few months, your team has accumulated a significant backlog of malware samples. These samples have been collected from various sources such as phishing emails, suspicious downloads, and through user reports.

To address this challenge, the organization has recently invested in enhancing its malware detection and threat intelligence capabilities by adopting YARA. Your task is to leverage your newly acquired YARA expertise to develop and refine YARA rules to better detect the threats and artifacts associated with these malware samples.

Challenge Directory:
05_Threat_Intelligence/YARA/Challenges/

Q1 First, run the strings command against HawkEye.exe.meow. What is the name of the JPG file that is referenced in the binary?
tmp1E2A.jpg

Q2 Use the name of the JPG file to create a YARA detection rule for any executable files that match this string identifier. Run it against HawkEye.exe.meow. What is the offset location of the matching detection?
0x1115d

Q3 What is the decimal conversion (base 10 equivalent) of this hexadecimal offset?
69981

Q4 Run the Rules/rule1.yar file against the entire Samples directory. Which file was detected?
RevengeRAT.exe.meow

Q5 Using the rule1.yar rule on the previously identified binary, how many individual string matches were detected?
4

Q6 Run the strings command against FreeYoutubeDownloader.exe.meow. What base domain name repeatedly appears in the output?
youtubedownloadernew.com

Q7 Use the previously identified base domain name, create a YARA detection rule for any executable files that match this string identifier. Run it against the FreeYoutubeDownloader.exe.meow binary. In order of appearance in the file, what are the hexadecimal offsets of the two matching detection strings?
0x358fa, 0x35983

Q8 Thinking back to the Pyramid of Pain, it would be quite trivial for an attacker to change the name of the program and any references to the previously identified domain name. However, consider the host artifacts that this tool inherently interacts with to function. Which interesting or unique registry key is found in the strings?
SOFTWARE\Borland\Delphi\RTL

Q9 Create a YARA rule only to detect this registry key, and run it recursively against the /YARA/Challenges/ folder. What other file(s) are detected? Only provide the name of the files, not the full path.

ZmtsO2d, c2RmZ2R

Q10 Create a YARA rule to detect any GIF89a files under 50 KB in size. Run it recursively against the /YARA/Challenges/ folder. Which file(s) are matched? Only provide the name of the files, not the full path.
vJ8AWaf

Q11 Create a YARA rule to detect any GIF89a files over 50 KB in size. Run it recursively against the /YARA/Challenges/ folder. Which file(s) are matched? Only provide the name of the files, not the full path.
fEbZizl

Q12 Research the file signature or magic bytes pertaining to OpenSSH private key files. What is the hexadecimal string (hex signature) that can be used to detect this signature?
2D 2D 2D 2D 2D 42 45 47 49 4E 20 4F 50 45 4E 53 53 48 20 50 52 49 56 41 54 45 20 4B 45 59 2D 2D 2D 2D 2D

Q13 Create a YARA rule to parse through the content of executables and detect any embedded OpenSSH private key files. Run it recursively against the /YARA/Challenges/ folder. Which file(s) are matched? Only provide the name of the files, not the full path.
VG8gQmF, Meoware.exe.meow

Challenge URL
https://challenges.malwarecube.com/#/c/f58d3f49-9bce-4e56-beb3-2488dfc3a31d